

Normal incidence x-ray standing wavefield absorption (NIXSW) of Quinacridone (QA) on the Ag(100) surface

Masterthesis

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1 Introduction

2 Literature

2.1 Quinacridone (QA)

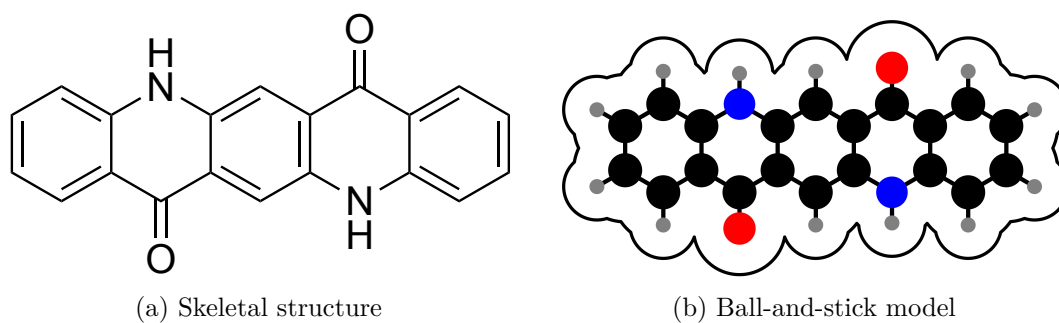


Figure 2.1: Molecular structure of quinacridone (QA).

2.2 Structures of Quinacridone (QA) on the Ag(100) surface

$$\mathbf{M}_\alpha = \begin{pmatrix} 2 & 1.25 \\ -3 & 4.80 \end{pmatrix},$$

$$\mathbf{M}_\beta = \begin{pmatrix} 4 & 3 \\ -5 & 3 \end{pmatrix},$$

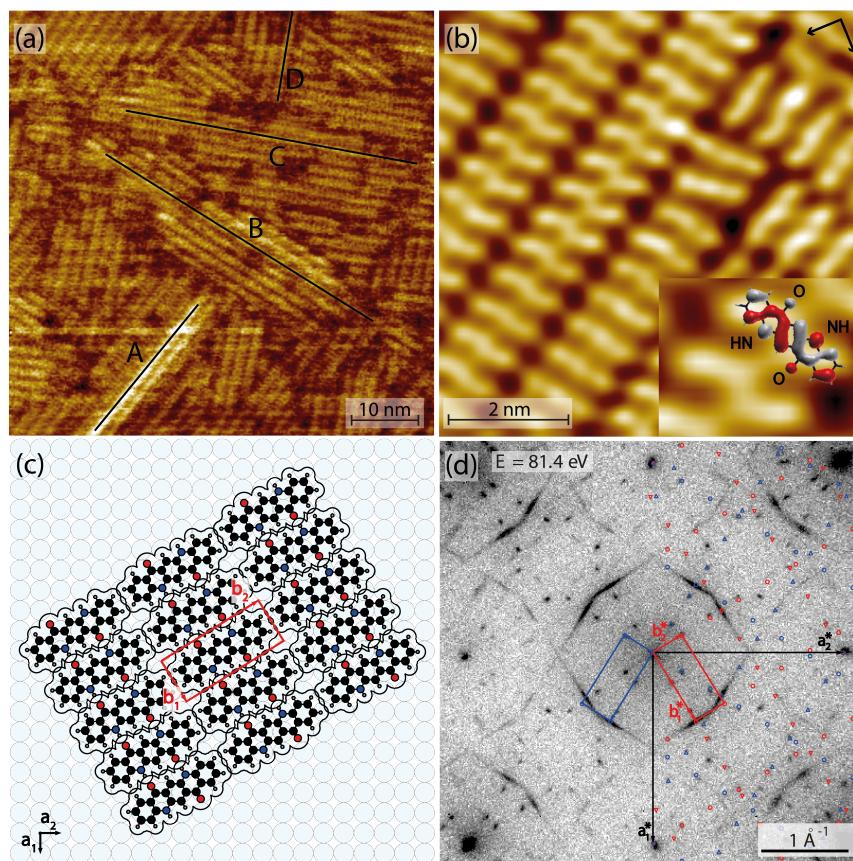


Figure 2.2: Current structure of complete monolayer (ML) of QA on Ag(100) in the α -phase. The image shows an overview scanning tunneling microscopy (STM) image (a), a close-up STM image of the molecular arrangement (b), the spot profile analysis low energy electron diffraction (SPA-LEED) pattern (c) and the structure model (d). The image is taken from reference [1].

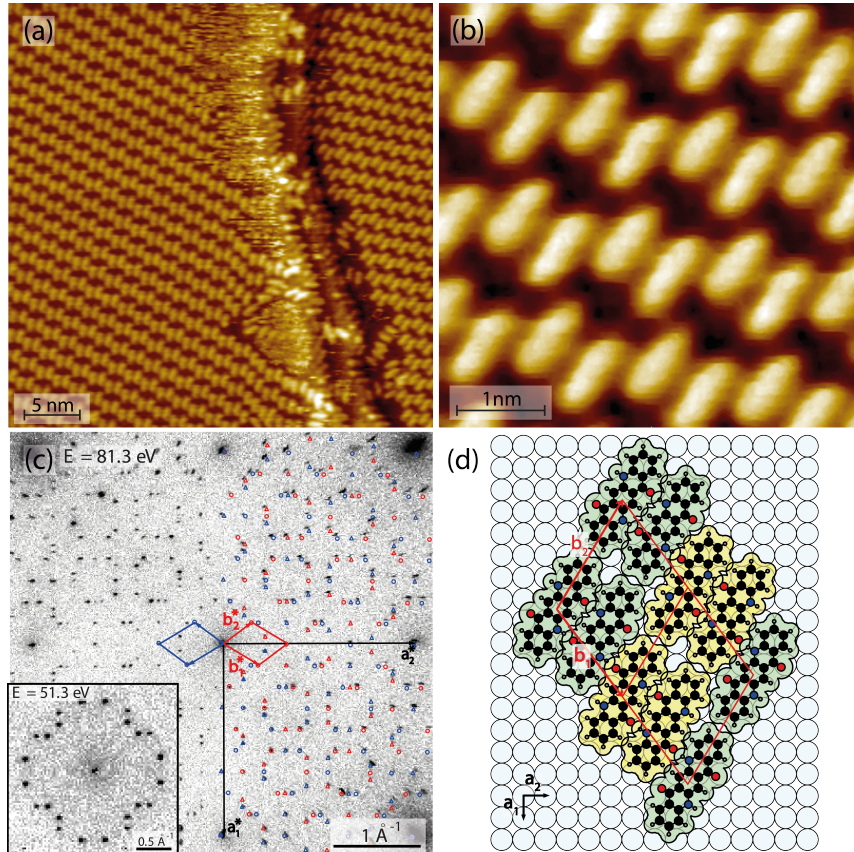


Figure 2.3: Current structure of complete ML of QA on Ag(100) in the β -phase. The image shows an overview STM image (a), a close-up STM image of the molecular arrangement (b), the SPA-LEED image (c) and the structure model (d). The image is taken from reference [1].

3 Theory

3.1 X-ray photoelectron spectroscopy (XPS)

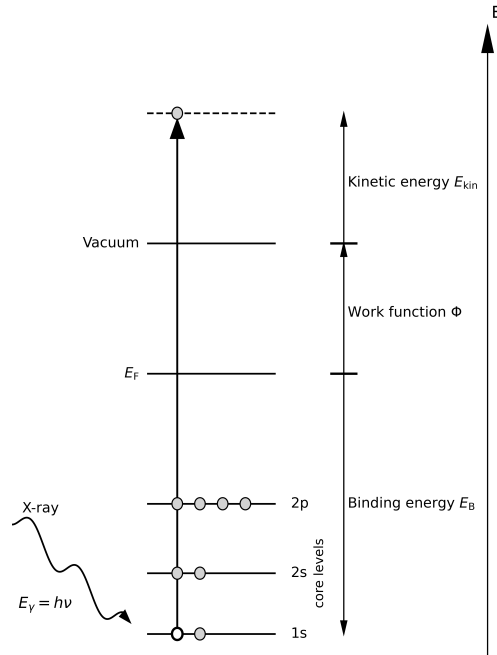


Figure 3.1: Schematic drawing of the photoemission process of an electron in the 1s orbital upon x-ray excitation. The image was adapted using reference [2].

$$E'_{kin} = E_{kin} + \phi_{sample}, \quad (3.1)$$

$$E_{kin} = E_\gamma - E_B - \phi_{sample}, \quad (3.2)$$

$$E_B = E_f(n-1) - E_i(n), \quad (3.3)$$

3.2 Normal Incidence X-ray Standing Wave (NIXSW)

4 Experimental

4.1 Material

4.1.1 The Ag(100) crystal surface

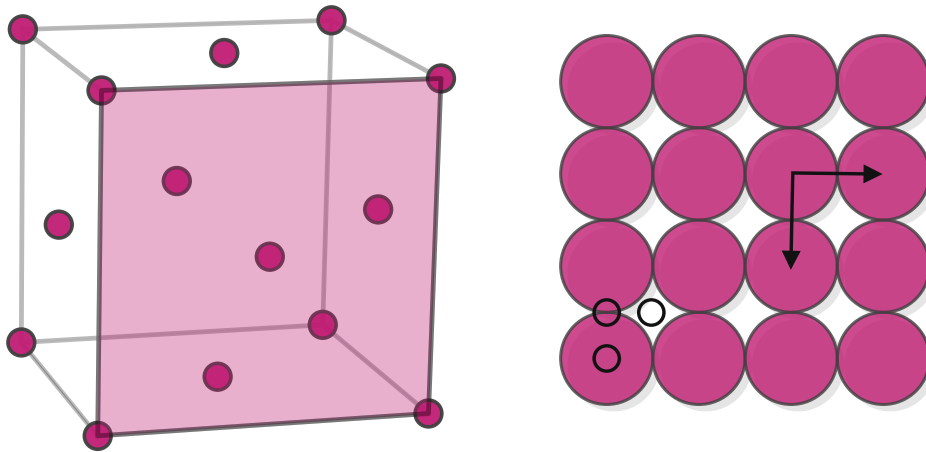


Figure 4.1: Unit cell of the silver crystal with marked (100) plane and the Ag(100) surface with the unit cell vectors (black arrows) and the distinct adsorption sites (black circles).

binding energy (BE)

4.2 Experimental setup

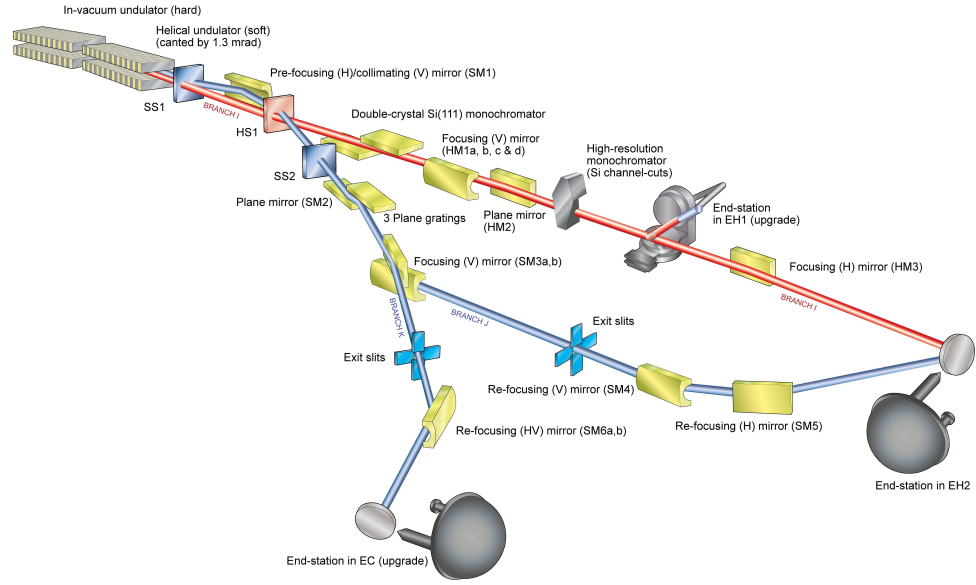


Figure 4.2: Schematic drawing of the I09 endstation of the Diamond Light Source in Didcot, UK. Picture taken from reference [3].

Table 4.1: Experimental parameters for XPS measurements at the I09 endstation of the Diamond Light Source in Didcot, UK.

	photon energy E_γ	pass energy E_{pass}	step width	number of frames
C1s	500 eV	20 eV	40 meV	17
O1s	630 eV	20 eV	40 meV	17
N1s	500 eV	20 eV	40 meV	17

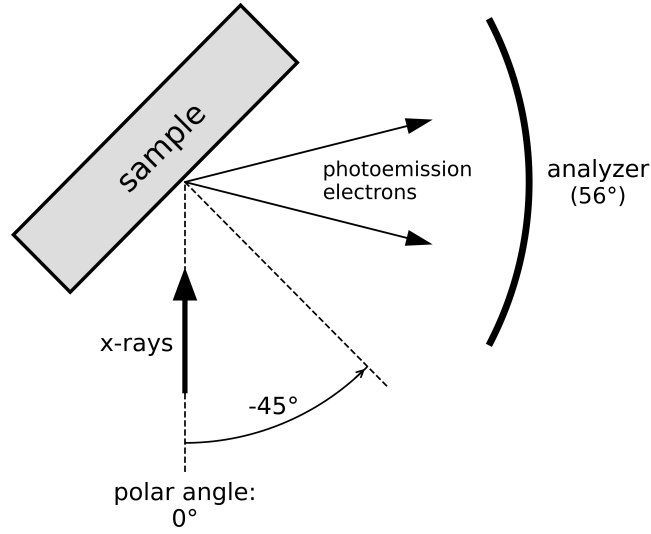


Figure 4.3: Top view of the sample geometries for the x-ray photoelectron spectroscopy (XPS) measurements at the I09 endstation of the Diamond Light Source in Didcot, UK. Illustrated as in reference [4].

4.3 Data processing

Table 4.2: Parameters used in CasaXPS^[5] for the different peaks of the XPS spectra of QA on Ag(100).

peak	line shape	area ratio
C_{arom}	LA(1,8,800)	10
C_{NH}	LA(1,8,800)	4
C_{CO}	LA(1,8,800)	4
C_{O}	LA(1,8,200)	2
O1	LA(1,8,400)	1 (for β -phase)
$O1_{\text{sat}}$	LA(1,1,400)	-
O2	LA(1,8,400)	1 (for β -phase)
$O2_{\text{sat}}$	LA(1,1,400)	-
N1	LA(1,8,400)	1 (for β -phase)
$N1_{\text{damage}}$	LA(1,8,400)	-
N2	LA(1,8,400)	1 (for β -phase)

5 XPS of QA on the Ag(100) surface

5.0.1 C1s XPS spectra

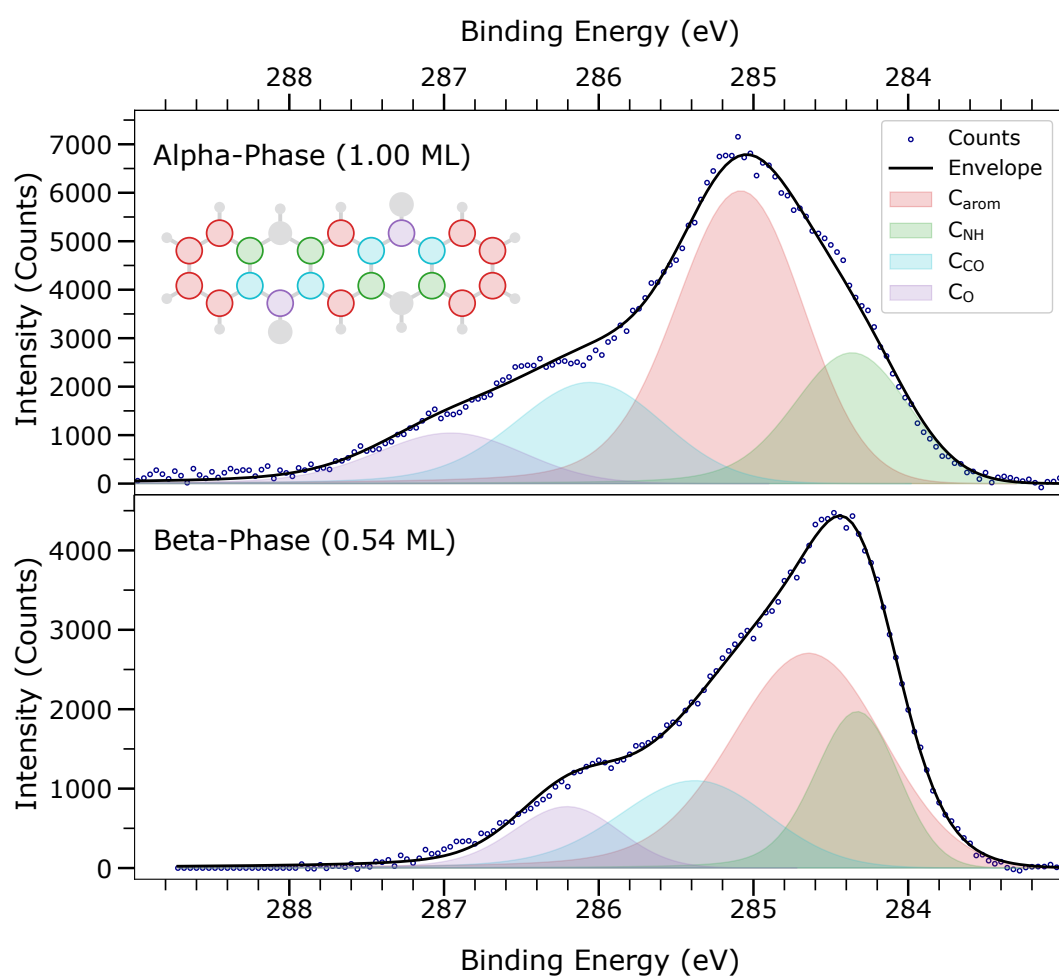


Figure 5.1: C1s XPS spectra of QA on the Ag(100) surface in the α -phase and in the β -phase. The different fitted peaks are shown in different colors.

Table 5.1

	peak	BE / eV	FWHM / eV	area ratio
α -phase	C _{arom}	285.00	0.95	10
	C _{NH}	284.29	0.85	4
	C _{CO}	285.96	1.1	4
	C _O	286.86	1.1	2
β -phase	C _{arom}	284.54	1.13	10
	C _{NH}	284.27	0.62	4
	C _{CO}	285.28	1.11	4
	C _O	285.13	0.79	2

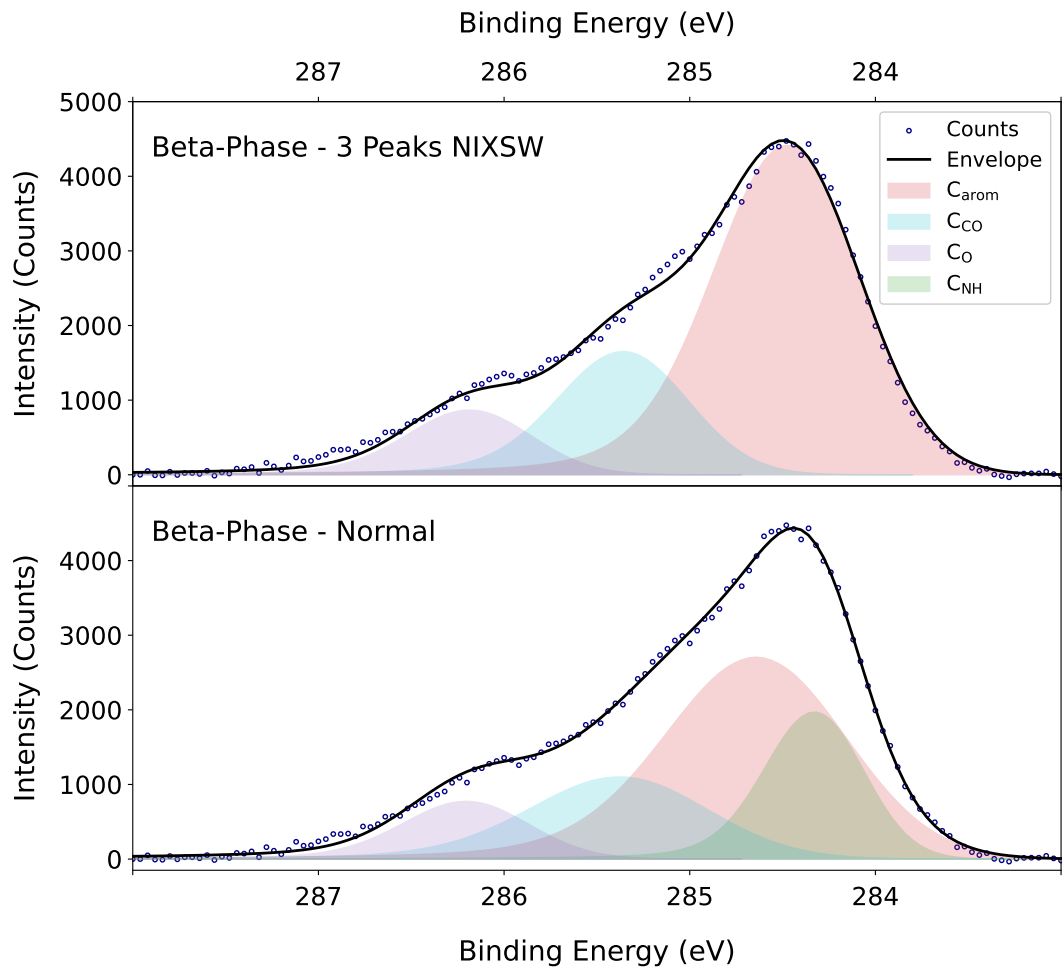


Figure 5.2: C1s XPS spectra of QA on the Ag(100) surface in the β -phase. The different fitted peaks are shown in different colors.

Table 5.2

	peak	BE / eV	area ratio	FWHM / eV
β -phase - 4 peaks	C _{arom}	284.54	1.13	10
	C _{NH}	284.27	0.62	4
	C _{CO}	285.28	1.11	4
	C _O	285.13	0.79	2
β -phase - 3 peaks	C _{arom} + C _{NH}	284.39	0.92	14
	C _{CO}	285.29	0.82	4
	C _O	286.12	0.78	2

5.0.2 O1s XPS spectra

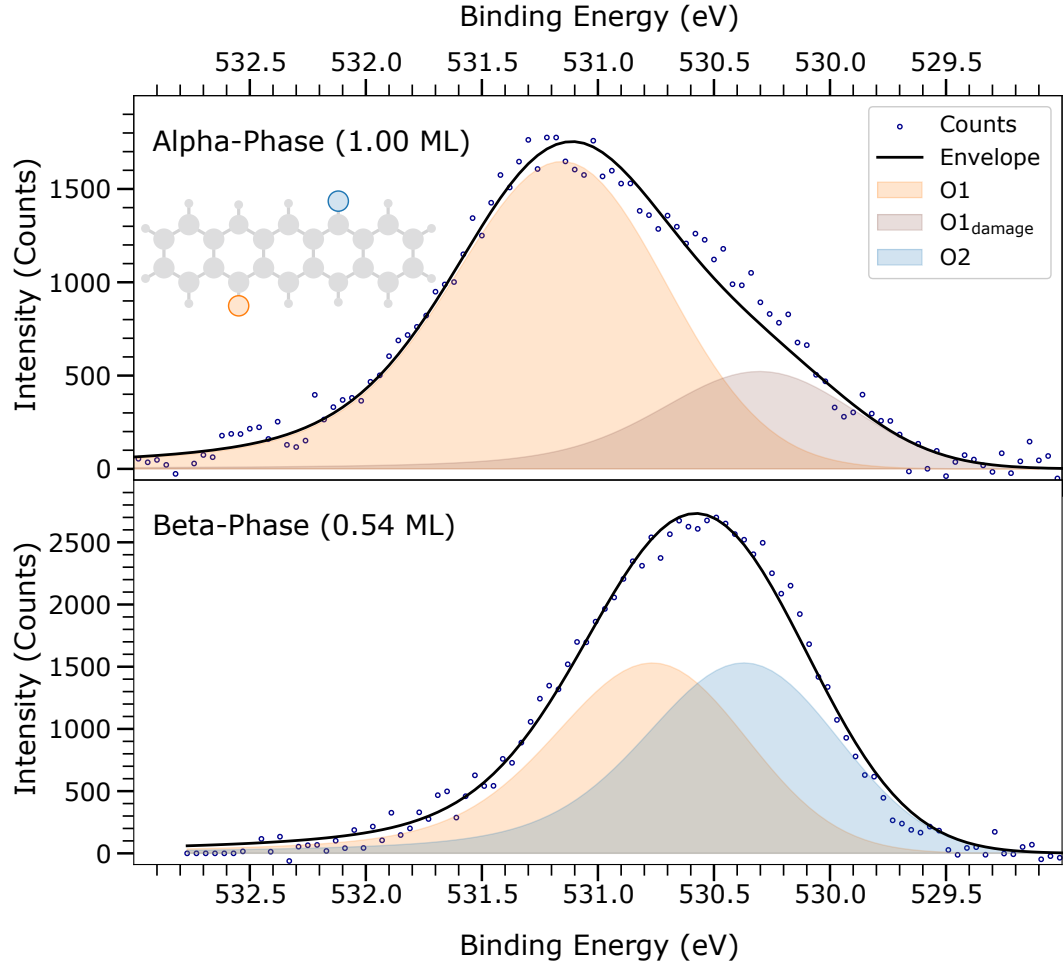


Figure 5.3: O1s XPS spectra of QA on the Ag(100) surface in the α -phase and in the β -phase. The different fitted peaks are shown in different colors.

Table 5.3

	peak	BE / eV	FWHM / eV	area ratio
α -phase	O1	531.02	1.05	1
	O1 _{damage}	530.18	0.95	-
β -phase	O1	530.64	0.95	1
	O2	530.25	0.95	1

5.0.3 N1s XPS spectra

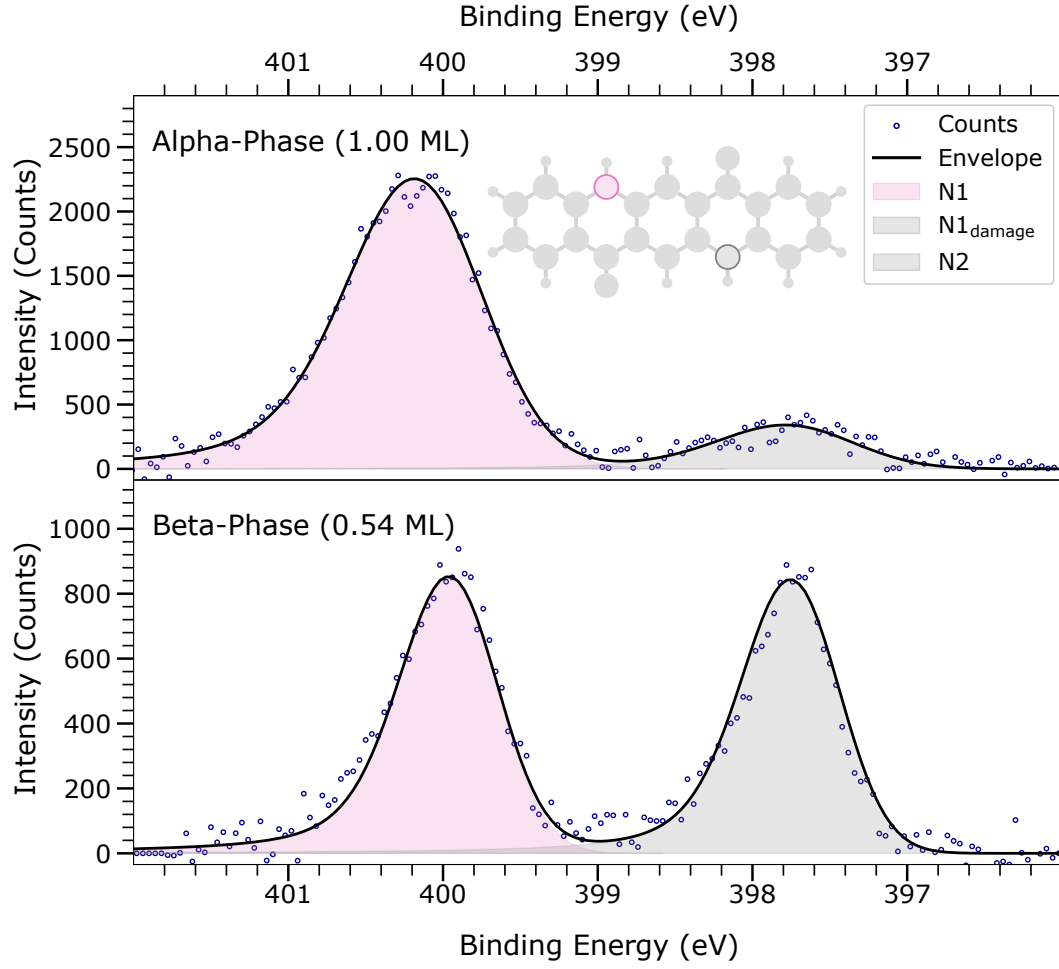


Figure 5.4: N1s XPS spectra of QA on the Ag(100) surface in the α -phase and in the β -phase. The different fitted peaks are shown in different colors.

Table 5.4

	peak	BE / eV	FWHM / eV	area ratio
α -phase	N1	400.05	1.02	1
	N1 _{damage}	397.66	1.02	-
β -phase	N1	399.87	0.73	1
	N2	397.66	0.73	1

6 NIXSW of QA on the Ag(100) surface

6.1 The α -phase

6.1.1 C1s NIXSW spectra

Table 6.1

	f_c	p_c	d (Å)
C _{arom}			
C _{NH}			
C _{CO}			
C _O			
C _{total}			

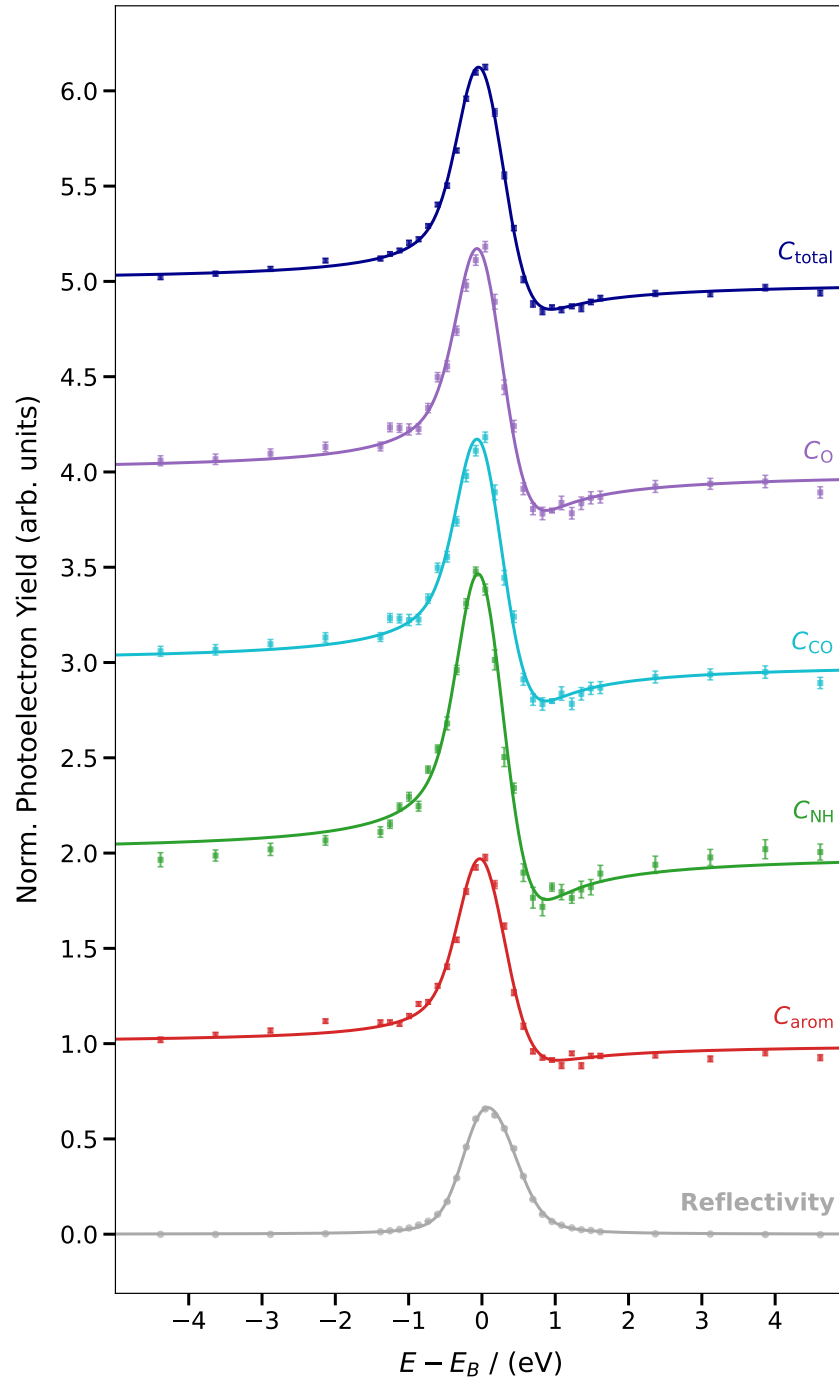


Figure 6.1: C1s NIXSW spectra of QA on the Ag(100) surface in the α -phase. The different fitted peaks are shown in different colors.

6.1.2 O1s NIXSW spectra

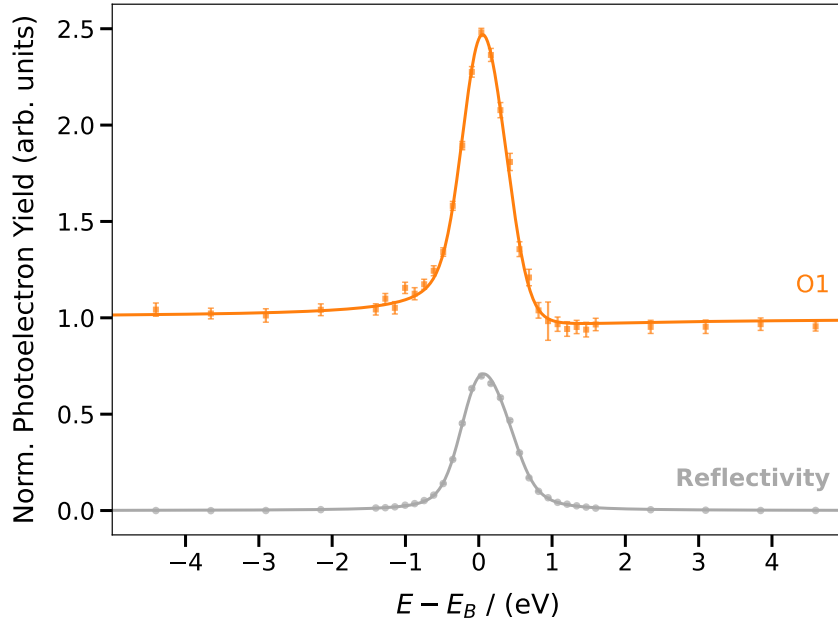


Figure 6.2: O1s NIXSW spectra of QA on the Ag(100) surface in the α -phase. The different fitted peaks are shown in different colors.

Table 6.2

	f_c	p_c	d (Å)
O1			
O1 _{damage}			
O _{total}			

6.1.3 N1s NIXSW spectra

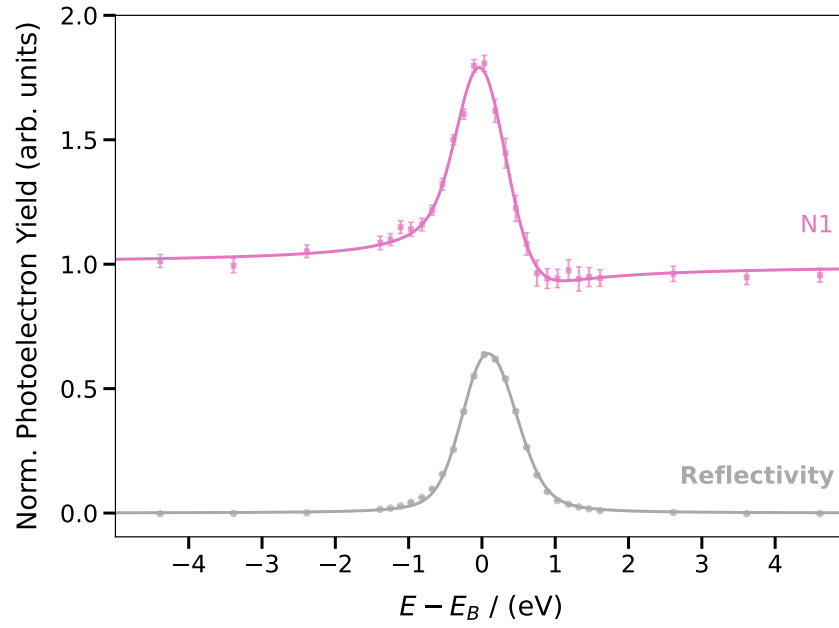


Figure 6.3: N1s NIXSW spectra of QA on the Ag(100) surface in the α -phase. The different fitted peaks are shown in different colors.

Table 6.3

	f_c	p_c	d (Å)
N1			
N1 _{damage}			
N _{total}			

6.2 The β -phase

6.2.1 C1s NIXSW spectra

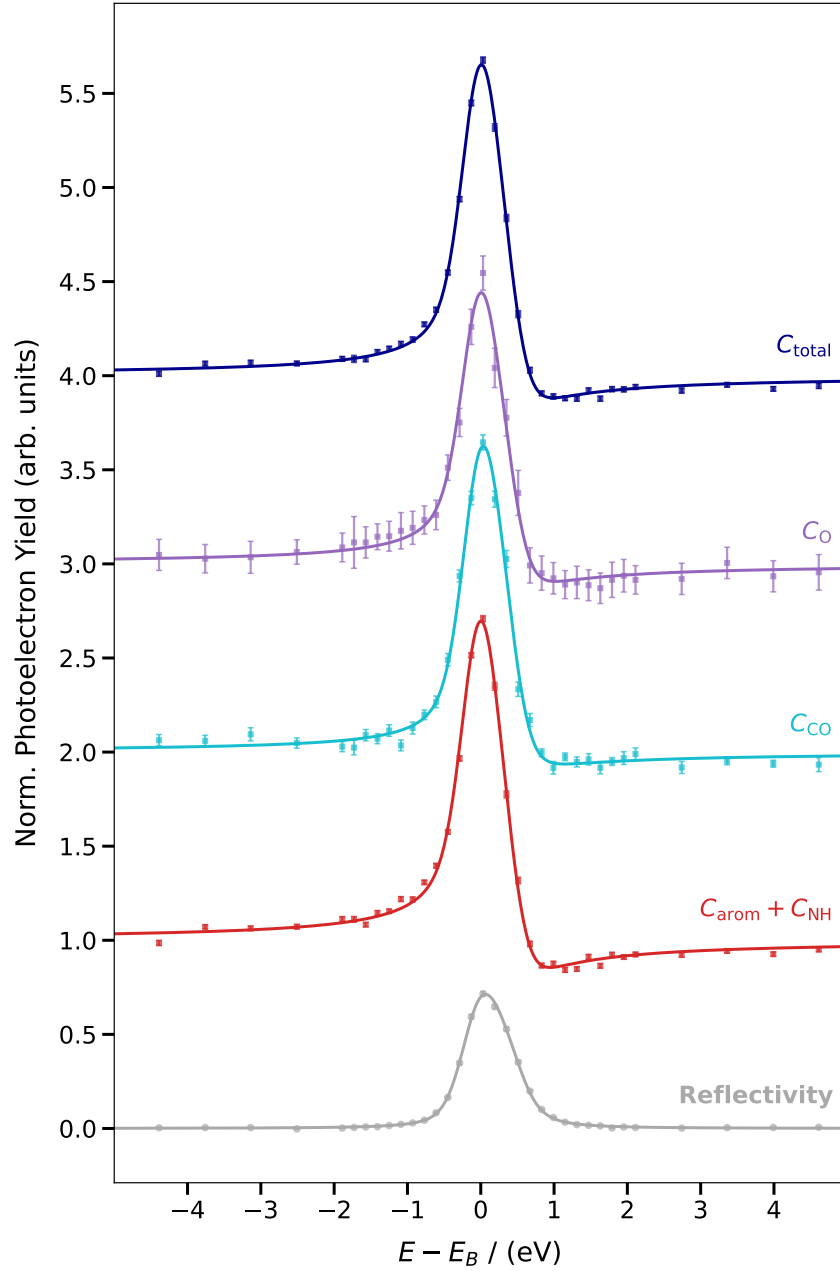


Figure 6.4: C1s NIXSW spectra of QA on the Ag(100) surface in the β -phase. The different fitted peaks are shown in different colors.

Table 6.4

	f_c	p_c	d (Å)
$C_{\text{arom}} + C_{\text{NH}}$			
C_{CO}			
C_{O}			
C_{total}			

6.2.2 O1s NIXSW spectra

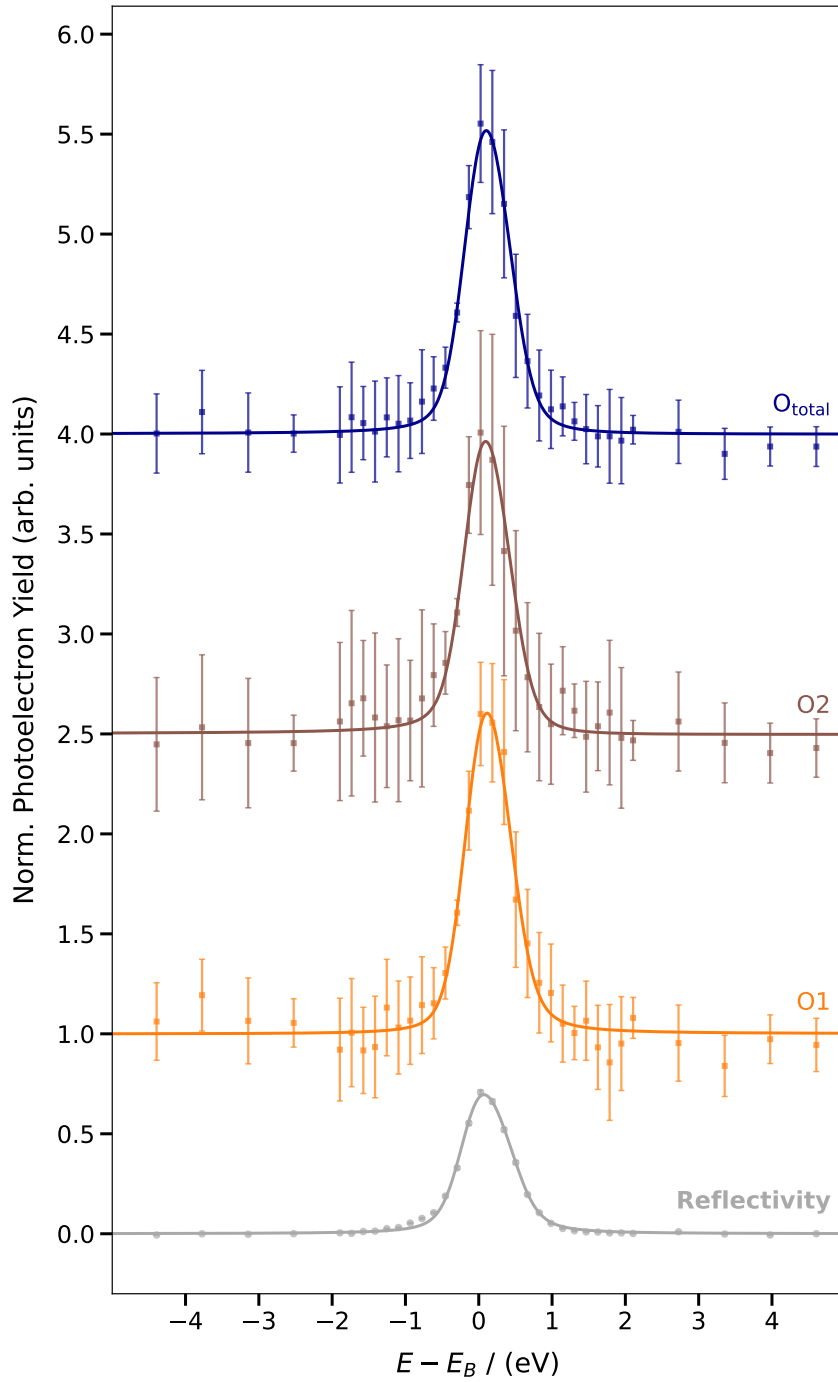


Figure 6.5: O1s NIXSW spectra of QA on the Ag(100) surface in the β -phase. The different fitted peaks are shown in different colors.

Table 6.5

	f_c	p_c	d (Å)
O1			
O2			
O _{total}			

6.2.3 N1s NIXSW spectra

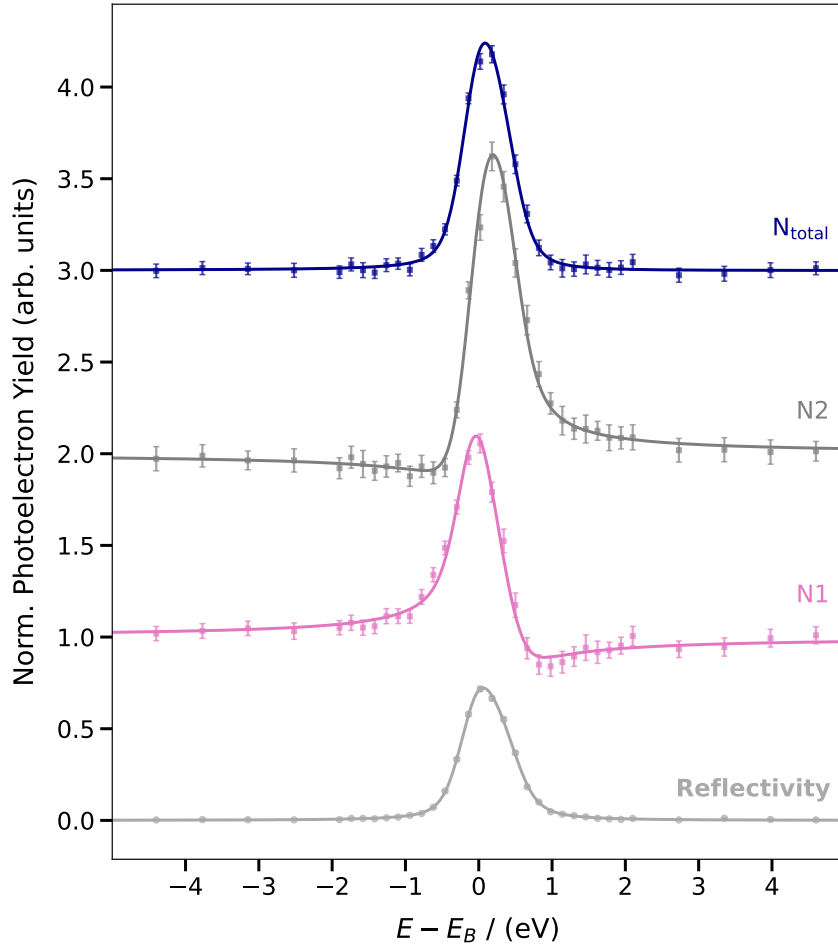


Figure 6.6: N1s NIXSW spectra of QA on the Ag(100) surface in the β -phase. The different fitted peaks are shown in different colors.

Table 6.6		
	f_c	p_c
N1		
N2		
N _{total}		

Overview of the α - and β -phase

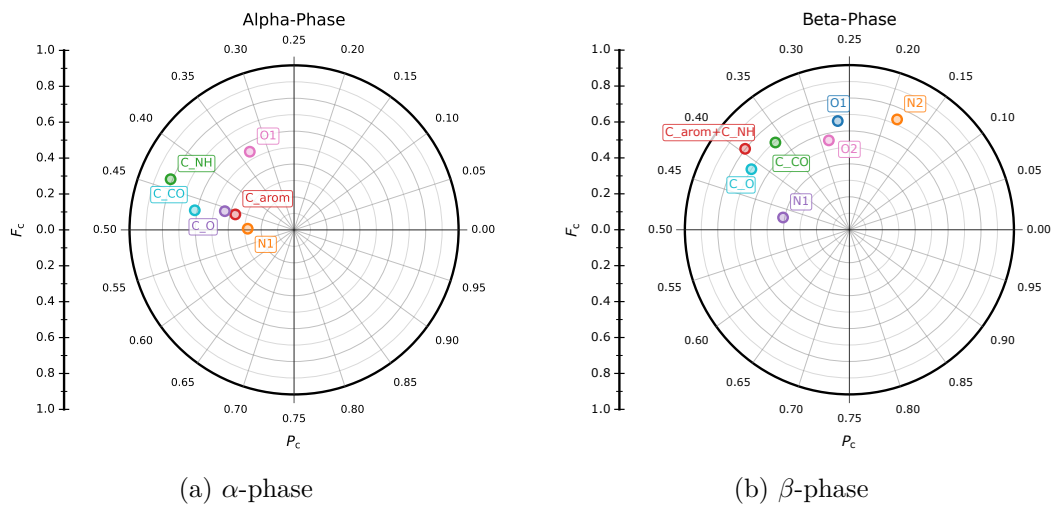
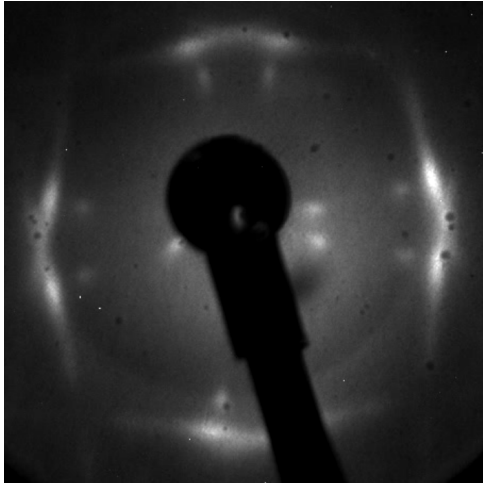


Figure 6.7: Overview of the Argand representations for the α - and β -phases.

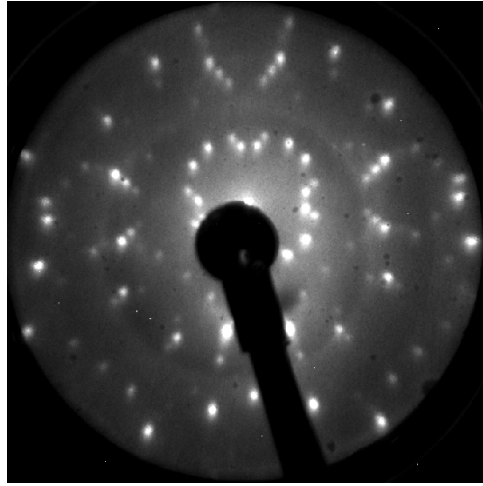
7 Discussion

8 Summary and Outlook

9 Appendix



(a) Point-on-line (POL) α -phase



(b) Commensurate β -phase

Figure 9.1: Low energy electron diffraction (LEED) patterns for the α - and the β -phase at 25.5 eV taken at the I09 beamline of the Diamond Light Source.

List of abbreviations

BE binding energy

fcc face-centered cubic

FWHM full width at half maximum

LEED low energy electron diffraction

ML monolayer

NIXSW normal incidence x-ray standing wavefield absorption

OFET organic field-effect transistor

OLED organic light-emitting diode

OPV organic photovoltaic

POL point-on-line

QA quinacridone

SPA-LEED spot profile analysis low energy electron diffraction

STM scanning tunneling microscopy

UHV ultra-high vacuum

XPS x-ray photoelectron spectroscopy

Bibliography

- [1] Hans Niklas Humberg. “One-Dimensional Aggregates of the Organic Dye Quinacridone on Metallic and Dielectric Surfaces.” Bonn: Rheinische Friedrich-Wilhelms-Universität Bonn, 2024.
- [2] Gary Attard and Colin Barnes. *Surfaces*. Oxford University Press, USA, June 11, 1998. 96 pp.
- [3] Diamond. *Diamond Light Source: I09*. June 2025. URL: <https://www.diamond.ac.uk/Instruments/Structures-and-Surfaces/I09.html>.
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- [5] Casa Software Ltd. *CasaXPS: Processing Software for XPS, AES, SIMS and More*. manual. 2022.